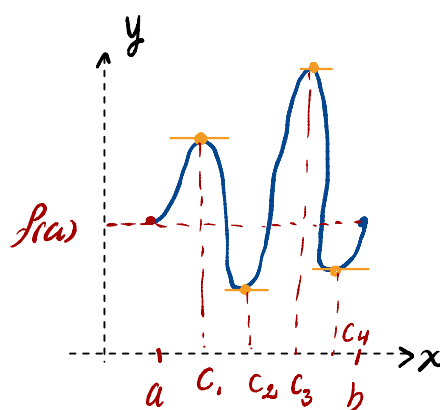
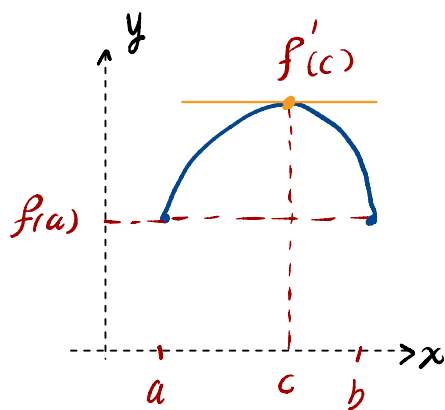
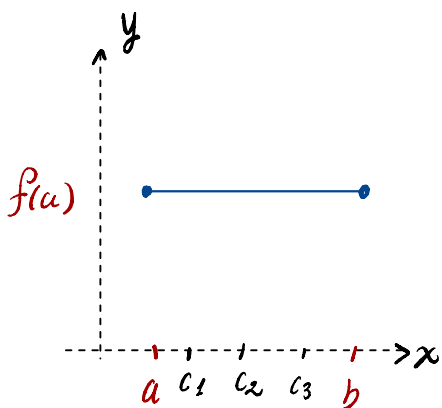


Section 3.2 (The mean Value Theorem).

The Rolle's Theorem: Let f be a function that satisfies the following conditions:

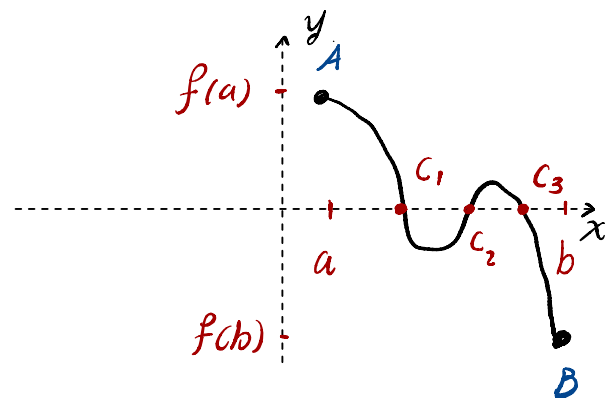
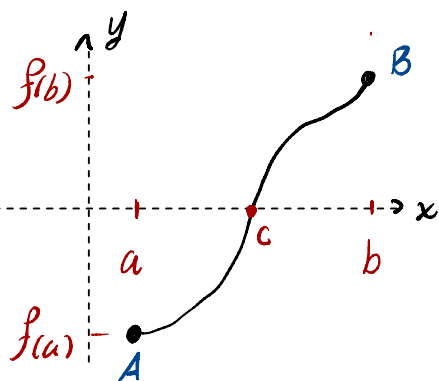
- i) f is continuous on a closed interval $[a, b]$.
- ii) f is differentiable on the open interval (a, b) .
- iii) $f(a) = f(b)$.

then, there is a number $c \in (a, b)$ such that $f'(c) = 0$.



Example: Prove that the equation $x^3 + x - 1 = 0$ has exactly one real solution.

Sln: Intermediate Value Theorem: If $f(x)$ is a continuous function on $[a, b]$ and $f(a) < 0$ and $f(b) > 0$ (or $f(a) > 0$ and $f(b) < 0$) then there is $c \in (a, b)$ s.t. $f(c) = 0$.



Let $f(x) = x^3 + x - 1$ on $[0, 1]$.

Notice that $f(x)$ is a continuous function on $[0, 1]$ and

$$f(0) = -1 \text{ but } f(1) = 1 + 1 - 1 = 1.$$

then there is $c \in (0, 1)$ s.t. $f(c) = 0$ or equivalently

$$c^3 + c - 1 = 0.$$

Rmk: We need to check that this is the only solution.

Now we need use the Rolle's theorem. By contradiction suppose that it had two solutions, a and b that is $f(a) = 0$ $f(b) = 0$.

Since $f(x)$ is a polynomial, $\begin{cases} \text{it's continuous on } [a, b] \\ \text{it's differentiable on } (a, b) \\ f(a) = 0 = f(b). \end{cases}$

then By Rolle's theorem there is $t \in (a, b)$ s.t. $f'(t) = 0$ But

$$f'(x) = 3x^2 + 1 \geq 1 \text{ for all } x \in (0, 1).$$

So $f'(x)$ can never be 0. This gives a contradiction. Therefore, the equation can't have two real solutions.

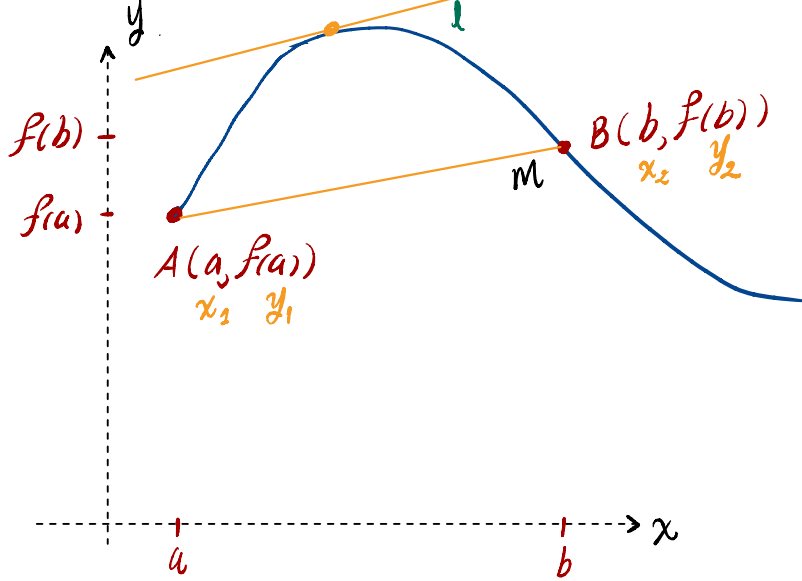
* The Mean Value theorem: Let f be a function that satisfies the following conditions:

1. f is a continuous function on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

then there is a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

or equivalently $f'(c) \cdot (b - a) = f(b) - f(a)$.



$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{f(b) - f(a)}{b - a}$$

then l is parallel to m .